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Inventor(s)	Yousefiani; Ali

Tuned multilayered material systems and methods for manufacturing

Abstract

A multilayered material system includes at least one of a liner sheet and a cellular core, and a multilayered composite joined to the at least one of a liner sheet and a cellular core. The multilayered composite includes hollow microspheres dispersed within a metallic matrix material.

Inventors:	**Yousefiani; Ali (Tustin, CA)**
Applicant:	**The Boeing Company** (Arlington, VA)
Family ID:	**76655764**
Assignee:	**The Boeing Company (Arlington, VA)**
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Primary Examiner: Collister; Elizabeth

Attorney, Agent or Firm: Walters & Wasylyna LLC

Background/Summary

PRIORITY (1) This application is a divisional of U.S. Ser. No. 16/733,498 filed on Jan. 3, 2020.

FIELD
(1) The present application relates to the field of multilayered materials and methods for manufacturing tuned multilayered material systems (TMMS), particularly tuned multilayered material systems for extreme environment hypersonic airframe structures, including the fuselage, wings, tails, control surfaces, leading edges, internal structure, air induction system, and thermal protection systems in general.
BACKGROUND
(2) Traditional materials with the ability to provide a path to manufacturable, durable, and rapidly deployable extreme environment hypersonic airframe structure, including the fuselage, wings, tails, control surfaces, leading edges, internal structure, and air induction system are expensive and require long fabrication cycles. To deliver affordable and robust airframe structures and thermal protection systems for future extreme environment applications, new technologies are required that can offer multilayered material systems tuned to locally meet stringent thermomechanical loading requirements on an airframe.
(3) Accordingly, those skilled in the art continue with research and development in the field of tuned multilayered material systems.
SUMMARY
(4) In one example, a graded multilayered composite comprises a metal matrix material having a first side and a second side opposite the first side. The graded multilayered composite also comprises a first layer of microspheres dispersed on the first side of the metal matrix material, and a second layer of microspheres dispersed on the second side of the metal matrix material.
(5) In another example, a graded multilayered material system comprises a non-graded multilayered composite. The graded multilayered material system also comprises at least one graded layer joined to the non-graded multilayered composite and selected from a graded metal

liner, a graded ceramic liner, a graded metal-ceramic hybrid liner, a graded metallic core, a graded cooling channel structure, and a graded environmental barrier coating.
(6) In yet another example, a method is provided for manufacturing a multilayered material system. The method comprises providing a graded multilayered composite, and joining at least one layer to the graded multilayered composite to provide the multilayered material system.
(7) In still another example, a method is provided for manufacturing a multilayered material system. The method comprises providing a non-graded multilayered composite, and joining at least one graded layer to the non-graded multilayered composite to provide the multilayered material system.
(8) In one example, a multilayered material system includes at least one of a liner sheet and a cellular core, and a multilayered composite (e.g., a multilayered metal matrix composite) joined to the at least one of a liner sheet and a cellular core. The multilayered composite includes hollow microspheres dispersed within a metallic matrix material.
(9) In another example, a method for manufacturing a multilayered composite includes providing a first layer of a first powder having first hollow microspheres dispersed therein, providing a second layer of a second powder adjacent the first layer of first powder, and heating the first layer of first powder and the second layer of second powder. The second layer of second powder has second hollow microspheres dispersed therein.
(10) In yet another example, a method for manufacturing a multilayered material system includes providing a first layer of a first powder having first hollow microspheres dispersed therein, providing a second layer of a second powder adjacent the first layer of first powder, sintering the first layer of first powder and the second layer of second powder, providing at least one of a liner sheet and a cellular core, and joining the first layer of sintered first powder with the at least one of a liner sheet and a cellular core. The second layer of the second powder has second hollow microspheres dispersed therein.
(11) Other examples of the disclosed multilayered material systems and methods of the present description will become apparent from the following detailed description, the accompanying drawings and the appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a cross-sectional view of an example of a graded multilayered composite according to the present description.
(2) FIG. **2**A is a cross-sectional view of the graded multilayered composite of FIG. **1** joined to a single-layer structure to form a graded multilayered material system.
(3) FIG. **2**B is a cross-sectional view of a non-graded multilayered composite joined to a graded single-layer structure to form a graded multilayered material system.
(4) FIG. **3**A is a cross-sectional view of the graded multilayered composite of FIG. **1** joined to a multiple-layer structure to form a graded multilayered material system.
(5) FIGS. **3**B-**3**E are cross-sectional views similar to FIG. **3**A, and show the graded multilayered composite of FIG. **1** joined to different multiple-layer structures to provide different graded multilayered material systems.
(6) FIG. **4** is a flow diagram representing a method for manufacturing a multilayered material system.
(7) FIG. **5** is a flow diagram representing a method for manufacturing a multilayered material system.
(8) FIG. **6** is a perspective view of a vehicle that includes a multilayered material system including a cellular sandwich panel and a multilayered composite joined to the cellular sandwich panel

according to the present description.
(9) FIG. **7** is a cross-sectional view of an example of the multilayered material system of FIG. **6**.
(10) FIG. **8** is a zoomed-in cross-sectional view of a portion of the multilayered material system of FIG. **7**.
(11) FIG. **9** is a cross-sectional view of another example of the multilayered material system of FIG. **6**.
(12) FIG. **10** is a zoomed-in cross-sectional view of a portion of the multilayered material system of FIG. **9**.
(13) FIG. **11** is flow diagram representing a method for manufacturing the multilayered composite of FIG. **6**.
(14) FIG. **12** is a flow diagram representing a method for manufacturing the multilayered material system of FIG. **6**.
(15) FIG. **13** is a flow diagram of an aircraft manufacturing and service methodology.
(16) FIG. **14** is a block diagram of an aircraft.
DETAILED DESCRIPTION
(17) FIG. **1** is a cross-sectional view of an example of a multilayered composite **1100** according to the present description. The multilayered composite **1100** is a graded multilayered composite that includes a metal matrix material **1110** having a first side **1112** and a second side **1114** opposite the first side **1112**. The graded multilayered composite **1100** also includes a first layer **1120** of microspheres dispersed on the first side **1112** of the metal matrix material **1110**, and a second layer **1122** of microspheres dispersed on the second side **1114** of the metal matrix material **1110**.
(18) The multilayered composite **1100** is graded based upon a combination of grading factors. As an example, a first portion of the multilayered composite **1100** may have a density that is different from a density of a second portion of the multilayered composite **1100**. As another example, the metal matrix material **1110** may comprise a compositionally-graded material, such as a hybrid titanium-based and nickel-based material system.
(19) As still another example, the microspheres of the first layer **1120** of microspheres may be spatially distributed relative to each other based upon a first spatial gradation, and the microspheres of the second layer **1122** of microspheres may be spatially distributed relative to each other based upon a second spatial gradation which is different from the first spatial gradation. In an example implementation, the first and second spatial gradations may be based upon number of microspheres. In another example implementation, the first and second spatial gradations may be based upon size of microspheres. Other grading factors and any combination of grading factors associated with the multilayered composite **1100** are possible.
(20) The graded multilayered composite **1100** further comprises a first buffer region **1140** defined between the first layer **1120** of microspheres and a first edge **1113** on the first side **1112** of the metal matrix material **1110**. The graded multilayered composite **1100** also comprises a second buffer region **1142** defined between the second layer **1122** of microspheres and a second edge **1115** on the second side **1114** of the metal matrix material **1110**. Each of the first buffer region **1140** and the second buffer region **1142** is substantially void of microspheres. The first buffer region **1140** and the second buffer region **1142** ensure that no partial microsphere is in the vicinity of the first edge **1113** and the second edge **1115**, which could result in a weak material stress point.
(21) FIG. **2**A is a cross-sectional view of the graded multilayered composite **1100** of FIG. **1** joined to a single-layer structure **1150** to form a graded multilayered material system **1200**. The single-layer structure **1150** may be selected from a metallic liner (e.g., a monolithic metal, metal alloy, metal matrix composite, intermetallic alloy, intermetallic matrix composite, complex concentrated alloy, or complex concentrated matrix composite), a ceramic liner (e.g. monolithic ceramic, ceramic matrix composite, or complex concentrated ceramic alloy), a metallic-ceramic hybrid liner, a metallic core, a cooling channel structure (which defines one or more cooling channels), and an environmental barrier coating. The single-layer structure **1150** may comprise a graded material

(e.g., a graded metallic or a graded ceramic or a graded hybrid).
(22) Although the graded multilayered material system **1200** of FIG. **2**A is formed using the graded multilayered composite **1100** of FIG. **1**, it is conceivable that a graded multilayered material system be formed using a substantially uniform (i.e., non-graded) multilayered composite and a graded single-layer structure such as shown in FIG. **2**B.
(23) As shown in FIG. **2**B, a multilayered material system **1250** comprises a non-graded multilayered composite **1260** (i.e., a substantially uniform multilayered composite) joined to a graded single-layered structure **1280**. The non-graded multilayered composite **1260** comprises a metal matrix material **1262** having a non-graded layer **1264** of microspheres dispersed in the metal matrix material **1262**. First and second buffer regions **1266**, **1268** are disposed on opposite sides of the non-graded layer **1264** of microspheres. The graded single-layered structure **1280** is joined to the first buffer region **1266**.
(24) In some examples, the non-graded multilayered composite **1260** comprises a substantially uniform composition of the metal matrix material **1262**. In some examples, the graded single-layered structure **1280** is selected from a monolithic or graded metallic liner (e.g., a metal, metal alloy, metal matrix composite, intermetallic alloy, intermetallic matrix composite, complex concentrated alloy, or complex concentrated matrix composite), a monolithic or graded ceramic liner (e.g., ceramic, ceramic matrix composite, or complex concentrated ceramic alloy), or a monolithic or graded metallic-ceramic hybrid liner, a graded metallic core, a graded cooling channel structure (which defines one or more cooling channels), and a graded environmental barrier coating.
(25) FIG. **3**A is a cross-sectional view of the graded multilayered composite **1100** of FIG. **1** joined to a multiple-layer structure to form a graded multilayered material system. Each of FIGS. **3**B-**3**E is a cross-sectional view similar to FIG. **3**A, and shows the graded multilayered composite **1100** of FIG. **1** joined to a different multiple-layer structure to provide a different graded multilayered material system. Each of the different multiple-layer structures may comprise a graded material structure.
(26) Although each of the graded multilayered material systems of FIGS. **3**A-**3**E is formed using the graded multilayered composite **1100** of FIG. **1**, it is conceivable that a multilayered material system be formed using a substantially uniform (i.e., non-graded) multilayered composite. For purposes of explanation, each of the graded multilayered material systems of FIGS. **3**A-**3**E will be described using the graded multilayered composite **1100** of FIG. **1**.
(27) As shown in graded multilayered material system **1300***a* of FIG. **3**A, graded multilayered composite **1100***a* is sandwiched between cooling channel structure **1310***a* (which defines one or more cooling channels **1311***a*) and first liner sheet **1320***a*. This sandwiched structure, in turn, is sandwiched between environmental barrier coating **1330***a* and cellular core **1340***a*. Second liner sheet **1350***a* is disposed on opposite side of cellular core **1340***a*. Environmental barrier coating **1330***a* may comprise a monolithic or graded metallic material (e.g., a metal, metal alloy, metal matrix composite, intermetallic alloy, intermetallic matrix composite, complex concentrated alloy, or complex concentrated matrix composite), a monolithic or graded ceramic material (e.g., ceramic, ceramic matrix composite, or complex concentrated ceramic alloy), or a monolithic or graded metallic-ceramic hybrid material. This environmental barrier coating **1330***a* can be provided for oxidation resistance, corrosion resistance, wear resistance, emissivity increase, etc.
(28) As shown in graded multilayered material system **1300***b* of FIG. **3**B, graded multilayered composite **1100***b* is sandwiched between cooling channel structure **1310***b* (which defines one or more cooling channels **1311***b*) and liner sheet **1320***b*. Environmental barrier coating **1330***b* is disposed on opposite side of cooling channel structure **1310***b*.
(29) As shown in graded multilayered material system **1300***c* of FIG. **3**C, graded multilayered composite **1100***c* is integrated with cooling channel structure **1310***c* (which defines one or more cooling channels **1311***c*). This integrated structure is sandwiched between first liner sheet **1320***c*

and second liner sheet **1350***c*. The first and second liner sheets **1320***c*, **1350***c* may comprise a monolithic or graded metallic material (e.g., a metal, metal alloy, metal matrix composite, intermetallic alloy, intermetallic matrix composite, complex concentrated alloy, or complex concentrated matrix composite), a monolithic or graded ceramic material (e.g., ceramic, ceramic matrix composite, or complex concentrated ceramic alloy), or a monolithic or graded metallic-ceramic hybrid material. Environmental barrier coating **1330***c* is disposed on opposite side of first liner sheet **1320***c*.

(30) As shown in graded multilayered material system **1300***f* of FIG. **3**D, graded multilayered composite **1100***f* is sandwiched between cooling channel structure **1310***f* (which defines one or more cooling channels **1311***f*) and environmental barrier coating **1330***f*. Cellular core **1340***f* is disposed on opposite side of cooling channel structure **1310***f*. Liner sheet **1320***f* is disposed on opposite side of cellular core **1340***f*.

(31) As shown in graded multilayered material system **1300***g* of FIG. **3**E, graded multilayered composite **1100***g* is integrated with cooling channel structure **1310***g* (which defines one or more cooling channels **1311***g*). This integrated structure is sandwiched first liner sheet **1320***g* and second liner sheet **1350***g*. This sandwiched structure, in turn, is sandwiched between first cellular core **1340***g* and second cellular core **1360***g*. This sandwiched structure, in turn, is sandwiched between third liner sheet **1370***g* and fourth liner sheet **1380***g*. Environmental barrier coating **1330***g* is disposed on opposite side of third liner sheet **1370***g*.

(32) In each of FIGS. **3**A-**3**E, it is conceivable that any number individual elements and any combination of the elements may be used to provide a graded multilayered material system. Moreover, it is conceivable that the multilayered composite may be integrated with any element.

(33) Referring to FIG. **4**, a flow diagram **1400** represents a method for manufacturing a multilayered material system. In block **1410**, a graded multilayered composite is provided. The process proceeds to block **1420** in which at least one layer is joined to the graded multilayered composite to provide the multilayered material system. The process then ends.

(34) In some examples, the at least one layer is selected from a monolithic or graded metallic liner (e.g., a metal, metal alloy, metal matrix composite, intermetallic alloy, intermetallic matrix composite, complex concentrated alloy, or complex concentrated matrix composite), a monolithic or graded ceramic liner (e.g., ceramic, ceramic matrix composite, or complex concentrated ceramic alloy), or a monolithic or graded metallic-ceramic hybrid liner, a monolithic or graded metallic core, a monolithic or graded cooling channel structure, or a monolithic or graded environmental barrier coating. For example, the monolithic or graded metallic liner comprises a metal, metal alloy, metal matrix composite, intermetallic alloy, intermetallic matrix composite, complex concentrated alloy, or complex concentrated matrix composite, and the monolithic or graded ceramic liner comprises ceramic, ceramic matrix composite, or complex concentrated ceramic alloy.

(35) Referring to FIG. **5**, a flow diagram **1500** represents a method for manufacturing a tuned multilayered material system. In block **1510**, a non-graded multilayered composite is provided. The process proceeds to block **1520** in which the at least one monolithic or graded layer is joined to the non-graded multilayered composite to provide the tuned multilayered material system. The process then ends.

(36) In some examples, the at least one graded layer is selected from a graded metal liner, a graded ceramic liner, a graded metal-ceramic hybrid liner, a graded metallic core, a graded cooling channel structure, and a graded environmental barrier coating. For example, the monolithic or graded metallic liner comprises a metal, metal alloy, metal matrix composite, intermetallic alloy, intermetallic matrix composite, complex concentrated alloy, or complex concentrated matrix composite, and the monolithic or graded ceramic liner comprises ceramic, ceramic matrix composite, or complex concentrated ceramic alloy.

(37) FIG. **6** is a perspective view of a structure **1** that includes a multilayered material system **10** according to the present description. The structure **1** is shown as an aircraft, such as a hypersonic

aircraft, but the structure **1** is not limited to vehicles and can include, for example, weapons, such as hypersonic weapons. The multilayered material system **10** can form an exterior surface of the structure **1** and can function as a thermal protection system for the structure **1**. It can also serve as other acreage skin structure, engine inlet structure, leading edge structure, control surface structure, thermo-mechanical isolator structure, or integrated thermal protection system for internal cold components.

(38) FIG. **7** is a cross-sectional view of an example of the multilayered material system **10** of FIG. **6**, and FIG. **8** is a zoomed-in cross-sectional view of a portion of the multilayered material system of FIG. **7**. As shown in FIGS. **7** and **8**, the tuned multilayered material system **10** includes a cellular sandwich panel **100** and a multilayered composite **200** joined to the cellular sandwich panel **100**, in which the multilayered composite **200** includes hollow microspheres **210** dispersed within a metallic matrix material **220**.

(39) FIG. **9** is a cross-sectional view of another example of the multilayered material system **10** of FIG. **6**, and FIG. **10** is a zoomed-in cross-sectional view of a portion of the tuned multilayered material system of FIG. **9**. As shown in FIGS. **9** and **10**, the multilayered material system **10** includes a cellular sandwich panel **100** and a multilayered composite **200** joined to the cellular sandwich panel **100**, in which the multilayered composite **200** includes a spatial distribution of hollow microspheres **210** dispersed within a metallic matrix material **220**.

(40) The multilayered material systems **10** of FIGS. **7** to **10** enable for the design of multifunctional and tunable structures that combine exceptional stiffness and strength-to-weight ratio with additional functional enhancements such as thermal protection and thermal management. The multilayered material system **10** includes two main constituents. First the cellular sandwich panel **100** can be optimized and tuned to meet specific extreme environment application requirements. Second, the multilayered composite **200** can be optimized and tuned to meet thermomechanical loading profile requirements. Further, the cellular sandwich panel **100** and the multilayered composite **200** can be joined together by a variety of methods to meet thermomechanical loading requirements.

(41) In an example, the cellular sandwich panel **100** includes a first liner sheet **110**, a second liner sheet **120**, and a cellular core **130** between the first liner sheet **110** and the second liner sheet **120**. The thickness of the cellular core **130** is typically greater than the thickness of the first liner sheet **110** and second liner sheet **120** and the density of the cellular core **130** is typically less than the density of the first liner sheet **110** and second liner sheet **120**. The stiffness of the first liner sheet **110** and second liner sheet **120** is typically greater than the stiffness of the cellular core **130**. By attaching the thinner but stiffer first liner sheet **110** and second liner sheet **120** to the lightweight by thicker cellular core **130**, the cellular sandwich panel **100** is provided with high stiffness and low overall density.

(42) The first liner sheet **110** can be formed from a variety of alloys, including but not limited to aluminum and aluminum alloys/metal matrix composites; titanium and titanium alloys/metal matrix composites; superalloys (including iron and iron alloys/metal matrix composites, nickel and nickel alloys/metal matrix composites, cobalt and cobalt alloys/metal matrix composites); refractory metals and alloys/metal matrix composites; copper and copper alloys/metal matrix composites; precious metals and alloys/metal matrix composites; zirconium and hafnium and their alloys/metal matrix composites; intermetallics; complex concentrated alloys/metal matrix composites (high entropy alloys/metal matrix composites, medium entropy alloys/metal matrix composites, multicomponent alloys/metal matrix composites). In an example, the first liner sheet **110** is formed from a titanium alloy. The first liner sheet **110** can be optimized and tuned to have a variety of thicknesses.

(43) The second liner sheet **120** can be formed from a variety of alloys, including but not limited to aluminum and aluminum alloys/metal matrix composites; titanium and titanium alloys/metal matrix composites; superalloys (including iron and iron alloys/metal matrix composites, nickel and

nickel alloys/metal matrix composites, cobalt and cobalt alloys/metal matrix composites); refractory metals and alloys/metal matrix composites; copper and copper alloys/metal matrix composites; precious metals and alloys/metal matrix composites; zirconium and hafnium and their alloys/metal matrix composites; intermetallics; complex concentrated alloys/metal matrix composites (high entropy alloys/metal matrix composites, medium entropy alloys/metal matrix composites, multicomponent alloys/metal matrix composites). In an example, the second liner sheet **120** is formed from a titanium alloy. The second liner sheet **120** can be optimized and tuned to have a variety of thicknesses.

(44) The cellular core **130** can be formed from a variety of alloys, including but not limited to aluminum and aluminum alloys/metal matrix composites; titanium and titanium alloys/metal matrix composites; superalloys (including iron and iron alloys/metal matrix composites, nickel and nickel alloys/metal matrix composites, cobalt and cobalt alloys/metal matrix composites); refractory metals and alloys/metal matrix composites; copper and copper alloys/metal matrix composites; precious metals and alloys/metal matrix composites; zirconium and hafnium and their alloys/metal matrix composites; intermetallics; complex concentrated alloys/metal matrix composites (high entropy alloys/metal matrix composites, medium entropy alloys/metal matrix composites, multicomponent alloys/metal matrix composites). In an example, the cellular core **130** is formed from a titanium alloy. The cellular core **130** can be optimized and tuned to have a variety of thicknesses.

(45) The cellular core **130** can be produced using a variety of additive manufacturing technologies, including melting processes, such as powder bed fusion or directed energy deposition; sintering processes, such as binder jetting, material extrusion, and material jetting; and solid state processes, such as additive friction stir processing, ultrasonic additive processing, cold spray, etc.

(46) The cellular core **130** can have a variety of architectures. In an example, cellular core **130** can have an open cellular architecture. In another example, the cellular core **130** can have a closed cellular architecture. In another example, the cellular core **130** can have a honeycomb architecture. The architecture of the cellular core **130** can be tuned and optimized to meet application requirements.

(47) The cellular core **130** can be bonded to the first liner sheet **110** and second liner sheet **120** by a variety of methods, such as by welding, brazing, fastening, diffusion bonding (with or without interlayer foils/coatings) or additive manufacturing.

(48) In an example, the cellular core **130** includes one or more third liner sheets. In another example, the cellular core **130** includes one or more third liner sheets **132** that are superplastically formed and are diffusion bonded to the first liner sheet **110** and the second liner sheet **120**. Superplastic forming and diffusion bonding (SPF/DB) is a technique for forming complex-shaped hollow cellular sandwich panels. It combines superplastic forming with diffusion bonding to create the cellular sandwich panels. Typically, three or more liner sheets are welded together at their edges, then heated within the confines of a female mold tool. At high temperatures, the three or more liner sheets become extremely malleable, i.e. superplastic. When in the superplastic state, an inert gas is injected between the three or more liner sheets to form the three or more liner sheets to the shape of the mold. Superplastic forming and diffusion bonding is useful for complex shapes. Thus, the architecture of the one or more third liner sheets **132** of the cellular core **130** can be tuned and optimized to meet a wide variety of application requirements. In the illustrated example, the cellular core **130** includes a double core structure having two third liner sheets **132**.

(49) The cellular sandwich panel **100** can provide a thermal protection gradient functionality. In an example, the melting point or thermal microstructural stability point of the first liner sheet **110** is greater than the melting point or thermal microstructural stability point of the second liner sheet **120**. In another example, the melting point or thermal microstructural stability point of the first liner sheet **110** is greater than the melting point or thermal microstructural stability point of the cellular core **130**. In yet another example, the melting point or thermal microstructural stability

point of the cellular core **130** is greater than the melting point or thermal microstructural stability point of the second liner sheet **120**. In yet another example, the melting point or thermal microstructural stability point of the first liner sheet **110** is greater than the melting point or thermal microstructural stability point of the cellular core **130**, which is greater than the melting point or thermal microstructural stability point of the second liner sheet **120**. By way of providing the above-described thermal protection gradient functionality, the cellular sandwich panel **100** has a hot side with higher resistance to high temperatures and a cold side with lower resistance to high temperatures.

(50) In addition, by relaxing the requirements for high resistance to high temperatures at the cold side, the cold side can be formed from materials having lower cost or superior properties, such as increased strength, increased damage tolerance, increased resistance to environmentally assisted cracking, increased formability, increased joinability or increased producibility, than the materials at the hot side. Accordingly, by way of example, the strength, damage tolerance, resistance to environmentally assisted cracking, formability, joinability or producibility of the second liner sheet **120** is greater than the strength, damage tolerance, resistance to environmentally assisted cracking, formability, joinability or producibility of the first liner sheet **110**. In another example, the strength, damage tolerance, resistance to environmentally assisted cracking, formability, joinability or producibility of the second liner sheet **120** is greater than the strength, damage tolerance, resistance to environmentally assisted cracking, formability, joinability or producibility of the cellular core **130**. In yet another example, the strength, damage tolerance, resistance to environmentally assisted cracking, formability, joinability or producibility of the cellular core **130** is greater than the strength, damage tolerance, resistance to environmentally assisted cracking, formability, joinability or producibility of the first liner sheet **110**. In yet another example, the strength, damage tolerance, resistance to environmentally assisted cracking, formability, joinability or producibility of the second liner sheet **120** is greater than the strength, damage tolerance, resistance to environmentally assisted cracking, formability, joinability or producibility of the cellular core **130**, which is greater than the strength, damage tolerance, resistance to environmentally assisted cracking, formability, joinability or producibility of the first liner sheet **110**. By way of providing the above-described thermal protection gradient functionality, the cellular sandwich panel **100** can have a hot side with higher resistance to high temperatures but lower strength, damage tolerance, resistance to environmentally assisted cracking, formability, joinability or producibility and a cold side with lower resistance to high temperatures but higher strength, damage tolerance, resistance to environmentally assisted cracking, formability, joinability or producibility.

(51) The first liner sheet **110** can include a first liner layer **112** proximate to the cellular core **130** and a second liner layer **114** proximate to the multilayered composite **200**. The first liner layer **112** and the second liner layer **114** can provide a thermal protection gradient functionality. In an example, a melting point or thermal microstructural stability point of the second liner layer **114** is greater than a melting point or thermal microstructural stability point of the first liner layer **112**. The first liner sheet **110** can further include third or further liner layers intermediate to the first liner layer **112** and the second liner layer **114**, which the third or further liner layers have a melting point or thermal microstructural stability points intermediate to the first liner layer **112** and the second liner layer **114**. By way of providing the above-described thermal protection gradient functionality of the first liner sheet **110**, the first liner sheet **110** has a hot side with higher resistance to high temperatures and a cold side with lower resistance to high temperatures.

(52) The first liner sheet **110** can provide for a compatibility with the multilayered composite **200**. In an aspect, the first liner layer **112** is compatible with the second liner layer **114**, which is compatible with the multilayered composite **200**, but the first liner layer **112** is incompatible or less compatible with the multilayered composite **200**. The first liner sheet **110** can further include third or further liner layers intermediate to the first liner layer **112** and the second liner layer **114**, in which the third or further liner layers are compatible with the first liner layer **112** and the second

liner layer **114** but the first liner layer **112** and second liner layer **114** are incompatible or less compatible with each other.

(53) In an example, a composition of the first liner layer **112** includes an element that is detrimental to the properties of the multilayered composite **200**, or a composition of the multilayered composite **200** includes an element that is detrimental the properties of the first liner layer **112**, and the second liner layer **114** excludes the detrimental element. Accordingly, the first liner sheet **110** can provide for an improved compatibility of the cellular sandwich panel **100** with the multilayered composite **200**.

(54) In another example, a temperature for processing the multilayered composite **200** exceeds the melting point or thermal microstructural stability point of the first liner layer **112** rendering the first liner layer **112** and the multilayered composite **200** incompatible, and the melting point or thermal microstructural stability point of the second liner layer **114** exceeds the temperature for processing the multilayered composite **200** rending the second liner layer **114** and the multilayered composite **200** more compatible. The temperature for processing the multilayered composite **200** can include, for example, a joining temperature, a sintering temperature, or a heat treatment temperature. Accordingly, the first liner sheet **110** can provide for an improved compatibility of the cellular sandwich panel **100** with the multilayered composite **200**.

(55) In yet another example, a coefficient of thermal expansion of the first liner layer **112** greatly varies from a coefficient of thermal expansion of the multilayered composite **200** and a coefficient of thermal expansion the second liner layer **114** varies less from the coefficient of thermal expansion of the multilayered composite **200**. Accordingly, the first liner sheet **110** can provide for an improved compatibility of the cellular sandwich panel **100** with the multilayered composite **200**.

(56) In addition, the first liner layer **112** can be formed from alloys having lower cost or superior properties, such as increased strength, damage tolerance, resistance to environmentally assisted cracking, formability, joinability or producibility. Accordingly, by way of example, the strength, damage tolerance, resistance to environmentally assisted cracking, formability, joinability or producibility of the first liner layer **112** is greater than the strength, damage tolerance, resistance to environmentally assisted cracking, formability, joinability or producibility of the second liner layer **114**. Thus, by providing the above-described compatibility of the cellular sandwich panel **100** with the multilayered composite **200**, the cellular sandwich panel **100** can be provided with a higher overall strength, damage tolerance, resistance to environmentally assisted cracking, formability, joinability or producibility while remaining compatible with the multilayered composite **200**.

(57) As previously mentioned, the multilayered composite **200** includes hollow microspheres **210** dispersed within a metallic matrix material **220**. The hollow microspheres **210** can provide the multilayered composite **200** with lightweight characteristics and insulative, conductive, and/or noise/impact attenuating properties. The metallic matrix material **220** can provide the multilayered composite **200** with durability and resistance to failure.

(58) The metallic matrix material **220** can be formed from a variety of materials. In an example, the metallic matrix material **220** is formed from at least one of an alloy material, including but not limited to aluminum and aluminum alloys/metal matrix composites; titanium and titanium alloys/metal matrix composites; superalloys (including iron and iron alloys/metal matrix composites, nickel and nickel alloys/metal matrix composites, cobalt and cobalt alloys/metal matrix composites); refractory metals and alloys/metal matrix composites; copper and copper alloys/metal matrix composites; precious metals and alloys/metal matrix composites; zirconium and hafnium and their alloys/metal matrix composites; intermetallics; complex concentrated alloys/metal matrix composites (high entropy alloys/metal matrix composites, medium entropy alloys/metal matrix composites, multicomponent alloys/metal matrix composites) and a ceramic material. By forming the metallic matrix material **220** from at least one of an alloy material and a ceramic material, the metallic matrix material **220** can be provided with resistance to high temperatures. In a specific example, the metallic matrix material **220** is formed from a nickel-based superalloy. In another

specific example, the metallic matrix material is formed from a titanium-based superalloy.
(59) The hollow microspheres **210** can be formed from a variety of materials. In an example, the hollow microspheres **210** are formed from a ceramic material. By forming the hollow microspheres **210** from a ceramic material, the hollow microspheres **210** can be provided with resistance to high temperatures and resistance against deformation to maintain their shape around the hollow interior thereof. In a specific example, the ceramic material is formed from yttria-stabilized zirconia or alumina-silica-iron glass. The architecture of the hollow microspheres **210** can be tuned and optimized to enable the multilayered composite **200** to meet application requirements. This architecture includes material, coating size, shell thickness, coating thickness, and type/material. In some implementations, the material of the hollow microspheres **210** is ceramic-based or metallic-based, and the size range is between 5 microns and 500 microns in diameter with average wall thickness between 2% to 30% of the diameter. In some implementations, the hollow microspheres **210** are coated with a coating made of metallic, ceramic, or hybrid metal-ceramic material combinations and having a coating thickness between 2 microns and 200 microns. The microspheres can also be solid. These example implementations tune the multilayered composite **200** for a particular application.
(60) In an example, the hollow microspheres **210** are included in the metallic matrix material **220** in a volume fraction in a range of between about 1 and 60 percent. Volume fraction of the hollow microspheres **210** is defined as the volume of all the hollow microspheres **210** within the metallic matrix material **220** divided by the total volume of the hollow microspheres **210** and the metallic matrix material **220**. A higher volume fraction of hollow microspheres **210** increases lightweight characteristics and insulative, conductive, and/or noise/impact attenuating properties of the multilayered composite **200**. A lower volume fraction of hollow microspheres **210** increases durability and resistance to failure of the multilayered composite **200**.
(61) In an example, the multilayered composite **200** includes a first layer **202** proximate to the first liner sheet **110** and a second layer **204** adjacent to first layer **202**. The first layer **202** has a first matrix **222** that includes first hollow microspheres **212**, and the second layer **204** has a second matrix **224** that includes second hollow microspheres **214**.
(62) The first layer **202** and second layer **204** can provide a thermal protection gradient functionality. In an example, a melting point or thermal microstructural stability point of the second matrix **224** is greater than a melting point or thermal microstructural stability point of the first matrix **222**. The multilayered composite **200** can further include third or further layers intermediate to the first layer **202** and the second layer **204**, in which the third or further layers have matrixes with melting point or thermal microstructural stability points that are intermediate to the melting point or thermal microstructural stability points of the first matrix **222** and second matrix **224**. By way of providing the above-described thermal protection gradient functionality of the multilayered composite **200**, the multilayered composite **200** has a hot side with higher resistance to high temperatures and a cold side with lower resistance to high temperatures.
(63) The multilayered composite **200** can provide for a compatibility with the cellular sandwich panel **100**. In an example, the first matrix **222** is compatible with the second liner layer **114** of the cellular sandwich panel **100**, but the second matrix **224** is incompatible or less compatible with the second liner layer **114** of the cellular sandwich panel **100**. The multilayered composite **200** can further include third or further layers intermediate to the first layer **202** and the second layer **204**, in which the third or further layers are compatible with the first layer **202** and the second layer **204** but the first layer **202** and second layer **204** are incompatible or less compatible with each other.
(64) In an example, a composition of the second matrix **224** includes an element that is detrimental to the properties of the second liner layer **114**, or a composition of the second liner layer **114** includes an element that is detrimental the properties of the second matrix **224**, and the second matrix **224** excludes the detrimental element. Accordingly, the multilayered composite **200** can provide for an improved compatibility with the cellular sandwich panel **100**.

(65) In another example, a temperature for processing the second matrix **224** exceeds the melting point or thermal microstructural stability point of the second liner layer **114** rendering the second matrix **224** and the second liner layer **114** incompatible, and the melting point or thermal microstructural stability point of the second liner layer **114** exceeds a temperature for processing the first matrix **222** rendering the first matrix **222** and the second liner layer **114** more compatible. The temperature for processing the first matrix **222** and the second matrix **224** can include, for example, a joining temperature, a sintering temperature, or a heat treatment temperature. Accordingly, the multilayered composite **200** can provide for an improved compatibility with the cellular sandwich panel **100**.

(66) In yet another example, a coefficient of thermal expansion of the second layer **204** greatly varies from a coefficient of thermal expansion of the second liner layer **114** and a coefficient of thermal expansion of the first layer **202** varies less from the coefficient of thermal expansion of the second liner layer **114**. Accordingly, the multilayered composite **200** can provide for an improved compatibility with the cellular sandwich panel **100**.

(67) In addition, the second matrix **224** can be formed from materials having lower cost or superior properties, such as higher resistance to high temperatures. Accordingly, by way of example, the melting point or thermal microstructural stability point of the second matrix **224** is greater than the melting point or thermal microstructural stability point of the first matrix **222**. Thus, by providing the above-described compatibility of the multilayered composite **200** with the cellular sandwich panel **100**, the multilayered composite **200** can be provided with a higher resistance to high temperatures while remaining compatible with the cellular sandwich panel **100**.

(68) As shown in FIGS. **9** and **10**, the first layer **202** and the second layer **204** provide a spatial distribution of the hollow microspheres **210**. Although FIGS. **9** and **10** show the first layer **202** and the second layer **204** providing a graded spatial distribution of the hollow microspheres **210**, it is conceivable that the first layer **202** and the second layer **204** provide a substantially uniform (i.e., non-graded) spatial distribution of the hollow microspheres **210**. For the purpose of explanation, only the graded spatial distribution of the hollow microspheres **210** will be described herein.

(69) As illustrated in FIGS. **9** and **10**, a volume fraction of the second hollow microspheres **214** within the second layer **204** is higher than a volume fraction of the first hollow microspheres **212** within the first layer **202**. By way of example, the volume fraction of the second hollow microspheres **214** within the second layer **204** is at least 5 percent greater than the volume fraction of the first hollow microspheres **212** within the first layer **202**, preferably at least 10 percent greater, more preferably at least 20 percent greater, even more preferably at least 50 percent greater, even more preferably at least 100 percent greater. Accordingly, the first layer **202** can have a higher durability and resistance to failure while the second layer **204** can have a lower overall density and higher insulative, conductive, and/or noise/impact attenuating properties. Additionally, the first layer **202** having a lower volume fraction of first hollow microspheres **212** can be more compatible for joining with the second liner layer **114** than the second layer **204** having a higher volume fraction of second hollow microspheres **214**. By way of a specific example, the first layer **202** has a volume fraction of about 10% first hollow microspheres **212**, and the second layer **204** has a volume fraction of about 45% second hollow microspheres **214**. The first liner sheet **110** can further include third or further liner layers intermediate to the first liner layer **112** and the second liner layer **114**, which the third or further liner layers have a melting point or thermal microstructural stability points intermediate to the first liner layer **112** and the second liner layer **114**. The multilayered composite **200** can further include third or further layers intermediate to the first layer **202** and the second layer **204**, in which the third or further layers having third or further hollow microspheres having different volume fractions of hollow microspheres.

(70) Although FIGS. **9** and **10** show the first layer **202** and second layer **204** as generally planar layers, in which the first layer **202** covers the surface of the cellular sandwich panel **100** and the second layer **204** covers the surface of the first layer **202**, other arrangements of the first layer **202**

and second layer **204** are included in the present description. For example, the first layer **202** and second layer **204** each cover adjacent portions of the cellular sandwich panel **100**. Accordingly, the first layer **202** can be more compatible for fastening with the second liner layer **114** of the cellular sandwich panel **100** than the second layer **204**. Thus, the first layer **202** can be positioned on the second liner layer **114** where fasteners connect the second liner layer **114** with the multilayered composite **200**.

(71) In another example, a composition of the second hollow microspheres **214** within the second layer **204** is different than a composition of the first hollow microspheres **212** within the first layer **202**. For example, a composition of the second hollow microspheres **214** are selected to provide higher insulative, conductive, and/or noise/impact attenuating properties than the insulative, conductive, and/or noise/impact attenuating properties of the composition of the first hollow microspheres **212**. Accordingly, the first layer **202** can have varying properties, such as insulative, conductive, and/or noise/impact attenuating properties, from the second layer **204**.

(72) In yet another example, a size of the second hollow microspheres **214** within the second layer **204** is different than a size of the first hollow microspheres **212** within the first layer **202**. Accordingly, the first layer **202** can have a varying insulative, conductive, and/or noise/impact attenuating properties from the second layer **204**.

(73) Referring back to FIGS. **7** to **10**, the multilayered material system **10** further includes a barrier coating **300** on a surface of the multilayered composite **200** to protect against environmental exposure and increase emissivity. The barrier coating **300** can have a variety of architectures, compositions, and thicknesses.

(74) The cellular sandwich panel **100** and a multilayered composite **200** are joined together by a variety of methods to form joint **400**, exemplary methods including welding, brazing, diffusion bonding, and fastening. In a specific example, the cellular sandwich panel **100** and a multilayered composite **200** are joined together to form joint **400** using a compositionally graded braze joint. In an example, the compositionally graded braze joint includes a first brazing layer adjacent to the cellular sandwich panel **100** and a second brazing layer adjacent to the multilayered composite **200**, wherein the first brazing layer has a coefficient of thermal expansion that is compatible with the cellular sandwich panel **100** and the second brazing layer has a coefficient of thermal expansion that is compatible with the multilayered composite **200**. In additional, the compositionally-graded braze joint can include third or additional brazing layer intermediate to the first brazing layer and second brazing layer having coefficients of thermal expansion that are intermediate to the coefficient of thermal expansion of the first brazing layer and the second brazing layer. Thus, the compositionally-graded braze joint can accommodate a coefficient of thermal expansion mismatch between the cellular sandwich panel **100** and a multilayered composite **200**.

(75) Although the multilayered material system **10** is illustrated in a planar configuration, the overall form of the multilayered material system **10** can vary. For example, curved or complex curved surfaces of acreage skin structure, engine inlet structure, leading edge structure, control surface structure, thermo-mechanical isolator structure, or integrated thermal protection systems for internal cold components can be formed from multilayered material system **10**.

(76) FIG. **11** is flow diagram representing a method **600** for manufacturing the multilayered composite **200** of FIG. **6**. The method **600** includes, at block **610**, providing a first layer of a first powder having first hollow microspheres **212** dispersed therein, and at block **620**, providing a second layer of a second powder adjacent the first layer of first powder, the second layer of second powder having second hollow microspheres **214** dispersed therein. The method **600** further includes, at block **630**, heating the first layer of first powder and the second layer of second powder. The heating can occur under various levels of sustained stress and for various durations.

(77) In an example, a melting point or thermal microstructural stability point of the second layer of second powder is greater than a melting point or thermal microstructural stability point of the first layer of first powder. According, a multilayered composite **200** can be provided with a thermal

protection gradient functionality as previously described above.

(78) In another example, a volume fraction of hollow microspheres within the second layer of second powder is higher than a volume fraction of hollow microspheres within the first layer of first powder. According, a multilayered composite **200** can be provided with a graded spatial distribution of the hollow microspheres **210** dispersed within a metallic matrix material **220**, as previously described.

(79) The first layer of first powder having first hollow microspheres **212** dispersed therein and the second layer of second powder having second hollow microspheres **214** dispersed therein may be provided in various ways. In an example, the first hollow microspheres **212** and second hollow microspheres **214** are pre-mixed into respective first powder and second powder. In another example, the first powder are provided as a first layer in a tool and then the first hollow microspheres **212** are placed within the first layer and the second powder are provided as a second layer in the tool and then the second hollow microspheres **214** are placed within the second layer.

(80) The second layer of second powder can be placed adjacent to the first layer of first powder by a variety of methods. In an example, the first layer of first powder is provided to a tool and then pressed with or without heat. Then the second layer of second powder is provided to the tool on the first layer and then pressed and heated together with the first layer of first powder. In another example, the first layer of first powder is provided to a tool and then an interlayer material, such an interlayer foil or interlayer mesh, is provided on the first layer. Then, the second layer of second powder is provided to the tool on the interlayer material and heated together with the first layer of first powder and the interlayer material. In yet another example, a mold is provided with an interlayer barrier separating a first compartment and second compartment. The first layer of first powder is provided to the first compartment and the second layer of second powder is provided to the second compartment, and then the first layer and second layer are heated together with the interlayer barrier. Thus, the first layer of first powder and second layer of second powder may be placed adjacent to each in various configurations.

(81) In an example, heating the first layer of first powder and the second layer of second powder includes heating the first layer of first powder and the second layer of second powder to a sintering temperature. The heating may include a consolidation process, such as hot isostatic pressing, spark plasma sintering, or cold isostatic pressing and sintering. In another example, heating the first layer of first powder and the second layer of second powder includes heating the first layer of first powder and the second layer of second powder to a heat treatment temperature.

(82) In an aspect, the first layer or the second layer are sintered, consolidated, or heat treated prior to a providing of the other of the first layer or the second layer. For example, the second layer of second powder can have a processing temperature, such as a sintering temperature, consolidation temperature, or heat treatment temperature, that is higher than a melting point or thermal microstructural stability point of the first layer of first powder. Thus, the second layer of second powder can be processed prior to providing of the first layer of first powder, then the first layer of first powder can be subject to processing, such as sintering, consolidation, or heat treatment. Accordingly, a multilayered composite **200** can be provided with a thermal protection gradient functionality as previously described above by separate processing of the first layer of first powder and second layer of second powder.

(83) FIG. **12** is a flow diagram representing a method **700** for manufacturing the multilayered material system **10** of FIG. **6**. The method **700** includes, at block **710**, providing a first layer of a first powder having first hollow microspheres **212** dispersed therein, at block **720**, providing a second layer of a second powder adjacent the first layer of first powder, the second layer of second powder having second hollow microspheres **214** dispersed therein, and, at block **730**, sintering the first layer of first powder and the second layer of second powder. The method **700** further includes, at block **740**, providing at least one of a liner sheet and a cellular core, and, at block **750**, joining the first layer of sintered first powder to the at least one of a liner sheet and cellular core. In some

implementations, the first layer of first powder and the second layer of second powder are sintered under necessary stress for necessary length of time.
(84) In an example, a melting point or thermal microstructural stability point of the second layer of second powder is greater than a melting point or thermal microstructural stability point of the first layer of first powder. According, a multilayered composite **200** can be provided with a thermal protection gradient functionality as previously described above.
(85) In another example, a volume fraction of hollow microspheres within the second layer of second powder is higher than a volume fraction of hollow microspheres within the first layer of first powder. According, a multilayered composite **200** can be provided with a graded spatial distribution of the hollow microspheres **210** dispersed within a metallic matrix material **220**, as previously described.
(86) The first layer of first powder having first hollow microspheres **212** dispersed therein and the second layer of second powder having second hollow microspheres **214** dispersed therein may be provided in various ways. In an example, the first hollow microspheres **212** and second hollow microspheres **214** are pre-mixed into respective first powder and second powder. In another example, the first powder are provided as a first layer in a tool and then the first hollow microspheres **212** are placed within the first layer and the second powder are provided as a second layer in the tool and then the second hollow microspheres **214** are placed within the second layer.
(87) The second layer of second powder can be placed adjacent to the first layer of first powder by a variety of methods. In an example, the first layer of first powder is provided to a tool and then pressed with or without heat. Then the second layer of second powder is provided to the tool on the first layer and then pressed and heated together with the first layer of first powder. In another example, the first layer of first powder is provided to a tool and then an interlayer material, such an interlayer foil or interlayer mesh, is provided on the first layer. Then, the second layer of second powder is provided to the tool on the interlayer material and heated together with the first layer of first powder and the interlayer material. In yet another example, a mold is provided with an interlayer barrier separating a first compartment and second compartment. The first layer of first powder is provided to the first compartment and the second layer of second powder is provided to the second compartment, and then the first layer and second layer are heated together with the interlayer barrier. Thus, the first layer of first powder and second layer of second powder may be placed adjacent to each in various configurations.
(88) In an example, the sintering the first layer of first powder and the second layer of second powder includes a consolidation process, such as hot isostatic pressing, spark plasma sintering, or cold isostatic pressing and sintering.
(89) In an aspect, the first layer or the second layer are sintered prior to a providing of the other of the first layer or the second layer. For example, the second layer of second powder can have a sintering temperature that is higher than a melting point or thermal microstructural stability point of the first layer of first powder. Thus, the second layer of second powder can be sintered prior to providing of the first layer of first powder, then the first layer of first powder can be subject to sintering. Accordingly, a multilayered composite **200** can be provided with a thermal protection gradient functionality as previously described above by separate processing of the first layer of first powder and second layer of second powder.
(90) The cellular sandwich panel **100** can take a variety of forms at previously described and may be formed according to a variety of methods. In an example, the step of providing the at least one of a liner sheet and cellular core includes, at block **742**, and a step of providing a first liner sheet **110**, at block **744**, a step of providing a second liner sheet **120**. The step of providing the at least one of a liner sheet and cellular core further includes, at block **746**, providing one or more third liner sheets **132** between the first liner sheet **110** and the second liner sheet **120**, and, at block **748**, superplastically forming and diffusion bonding the one or more third liner sheets to the first liner sheet and the second liner sheet.

(91) The step of joining the first layer of sintered first powder to the at least one of a liner sheet and cellular core may be performed by a variety of methods. Exemplary methods include welding, brazing, diffusion bonding, and fastening. In a specific example, the at least one of a liner sheet and cellular core and a multilayered composite **200** are joined together to form joint **400** using a compositionally-graded braze joint. In an example, the step of joining the first layer of sintered first powder to the at least one of a liner sheet and cellular core includes providing a first brazing layer adjacent to the at least one of a liner sheet and cellular core and a second brazing layer adjacent to the multilayered composite **200**. The first brazing layer can have a coefficient of thermal expansion that is compatible with the cellular sandwich panel **100** and the second brazing layer can have a coefficient of thermal expansion that is compatible with the multilayered composite **200**. In additional, the compositionally-graded braze joint can include third or additional brazing layer intermediate to the first brazing layer and second brazing layer having coefficients of thermal expansion that are positioned intermediate to the coefficient of thermal expansion of the first brazing layer and the second brazing layer. Thus, the compositionally-graded braze joint can accommodate a coefficient of thermal expansion mismatch between the at least one of a liner sheet and cellular core and a multilayered composite **200**.
(92) The above description describes numerous materials. It should be understood that "metal/metallic" includes "metals and metal matrix composites"; "ceramic" includes "ceramics and ceramic matrix composites"; and "hybrid metal-ceramic" includes "metal-ceramic hybrid and metal matrix composite/ceramic matrix composite hybrid". Also, metallic bases include aluminum and aluminum alloys/metal matrix composites; titanium and titanium alloys/metal matrix composites; superalloys (including iron and iron alloys/metal matrix composites, nickel and nickel alloys/metal matrix composites, cobalt and cobalt alloys/metal matrix composites); refractory metals and alloys/metal matrix composites; copper and copper alloys/metal matrix composites; precious metals and alloys/metal matrix composites; zirconium and hafnium and their alloys/metal matrix composites; intermetallics; complex concentrated alloys/metal matrix composites (high entropy alloys/metal matrix composites, medium entropy alloys/metal matrix composites, multicomponent alloys/metal matrix composites).
(93) It should be apparent that each of the graded multilayered composite **1100** of FIG. **1**, the graded multilayered material system **1200** of FIG. **2**, the multilayered material systems **1300*a*-1300*g*** of FIGS. **3**A-**3**E, and the multilayered material systems **10** of FIGS. **7-10** disclosed herein comprises either a tuned multilayered composite or a multilayered material system that can operate under stringent thermomechanical loading requirements, such as on an aircraft.
(94) Examples of the present disclosure may be described in the context of an aircraft manufacturing and service method **1000**, as shown in FIG. **13**, and an aircraft **1002**, as shown in FIG. **14**. During pre-production, the aircraft manufacturing and service method **1000** may include specification and design **1004** of the aircraft **1002** and material procurement **1006**. During production, component/subassembly manufacturing **1008** and system integration **1010** of the aircraft **1002** takes place. Thereafter, the aircraft **1002** may go through certification and delivery **1012** in order to be placed in service **1014**. While in service by a customer, the aircraft **1002** is scheduled for routine maintenance and service **1016**, which may also include modification, reconfiguration, refurbishment and the like.
(95) Each of the processes of method **1000** may be performed or carried out by a system integrator, a third party, and/or an operator (e.g., a customer). For the purposes of this description, a system integrator may include without limitation any number of aircraft manufacturers and major-system subcontractors; a third party may include without limitation any number of venders, subcontractors, and suppliers; and an operator may be an airline, leasing company, military entity, service organization, and so on.
(96) Any combination of the graded multilayered composite **1100** of FIG. **1**, the graded multilayered material system **1200** of FIG. **2**, the multilayered material systems **1300*a*-1300*g*** of

FIGS. **3**A-**3**E, and the multilayered material systems **10** of FIGS. **7-10** may be employed during any one or more of the stages of the aircraft manufacturing and service method **1000**, including specification and design **1004** of the aircraft **1002**, material procurement **1006**, component/subassembly manufacturing **1008**, system integration **1010**, certification and delivery **1012**, placing the aircraft in service **1014**, and routine maintenance and service **1016**.
(97) As shown in FIG. **14**, the aircraft **1002** produced by example method **1000** may include an airframe **1018** with a plurality of systems **1020** and an interior **1022**. Examples of the plurality of systems **1020** may include one or more of a propulsion system **1024**, an electrical system **1026**, a hydraulic system **1028**, and an environmental system **1030**. Any number of other systems may be included. The multilayered material system **10** of the present disclosure may be employed for any of the systems of the aircraft **1002**.
(98) Although various examples of the disclosed multilayered material systems and multilayered composites have been shown and described, modifications may occur to those skilled in the art upon reading the specification. The present application includes such modifications and is limited only by the scope of the claims.

Claims

1. A method for manufacturing a multilayered composite, comprising: providing a first layer of a first powder having first hollow microspheres dispersed therein; providing a second layer of a second powder over the first layer of the first powder, the second layer of the second powder having second hollow microspheres dispersed therein; and heating the first layer of the first powder and the second layer of the second powder, wherein at least one of: (i) the first layer of the first powder comprises a first metallic powder composition, the second layer of the second powder comprises a second metallic powder composition, and the second metallic powder composition is compositionally different from the first metallic powder composition; and (ii) a thermal microstructural stability point of the second layer of the second powder is greater than a thermal microstructural stability point of the first layer of the first powder.
2. The method of claim 1, wherein the providing the first layer of the first powder having first hollow microspheres dispersed therein comprises providing the first layer of the first powder having ceramic hollow microspheres dispersed therein.
3. The method of claim 1, wherein the heating comprises sintering the first layer of the first powder and the second layer of the second powder.
4. The method of claim 1, wherein the second layer of the second powder is adjacent to the first layer of the first powder.
5. The method of claim 1, wherein a melting point of the second layer of the second powder is greater than a melting point of the first layer of the first powder.
6. The method of claim 1, wherein a volume fraction of the second hollow microspheres within the second layer of the second powder is higher than a volume fraction of the first hollow microspheres within the first layer of the first powder.
7. The method of claim 6, wherein the volume fraction of the second hollow microspheres within the second layer of the second powder is at least 10 percent greater than the volume fraction of the first hollow microspheres within the first layer of the first powder.
8. The method of claim 6, wherein the volume fraction of the second hollow microspheres within the second layer of the second powder is at least 20 percent greater than the volume fraction of the first hollow microspheres within the first layer of the first powder.
9. The method of claim 6, wherein the volume fraction of the second hollow microspheres within the second layer of the second powder is at least 50 percent greater than the volume fraction of the first hollow microspheres within the first layer of the first powder.
10. The method of claim 1, further comprising: after the heating the first layer of the first powder

and the second layer of the second powder, joining the first layer of the first powder with a liner sheet.
11. The method of claim 10, wherein the joining comprises at least one of welding, brazing, and diffusion bonding.
12. The method of claim 1, further comprising: after the heating the first layer of the first powder and the second layer of the second powder, joining the first layer of the first powder with a cellular sandwich panel comprising a cellular core and at least one liner sheet.
13. The method of claim 12, wherein the joining comprises at least one of welding, brazing, and diffusion bonding.
14. The method of claim 12, wherein a melting point of the cellular core is greater than a melting point of the liner sheet.
15. The method of claim 12, further comprising: after the heating the first layer of the first powder and the second layer of the second powder, applying a barrier coating over the second layer of the second powder.
16. The method of claim 1, further comprising: providing at least one of a liner sheet and a cellular core; and after the heating, joining the multilayered composite with the at least one of the liner sheet and the cellular core.
17. A method for manufacturing a multilayered composite, comprising: providing a first layer of a first powder having first hollow microspheres dispersed therein, the first layer of the first powder comprising a first metallic powder composition; providing a second layer of a second powder over the first layer of the first powder, the second layer of the second powder having second hollow microspheres dispersed therein, the second layer of the second powder comprising a second metallic powder composition, wherein the second metallic powder composition is compositionally different from the first metallic powder composition; and heating the first layer of the first powder and the second layer of the second powder.
18. The method of claim 17, further comprising: providing at least one of a liner sheet and a cellular core; and after the heating, joining the multilayered composite with the at least one of the liner sheet and the cellular core.
19. A method for manufacturing a multilayered composite, comprising: providing a first layer of a first powder having first hollow microspheres dispersed therein; providing a second layer of a second powder over the first layer of the first powder, the second layer of the second powder having second hollow microspheres dispersed therein; and heating the first layer of the first powder and the second layer of the second powder, wherein a thermal microstructural stability point of the second layer of the second powder is greater than a thermal microstructural stability point of the first layer of the first powder.
20. The method of claim 19, further comprising: providing at least one of a liner sheet and a cellular core; and after the heating, joining the multilayered composite with the at least one of the liner sheet and the cellular core.